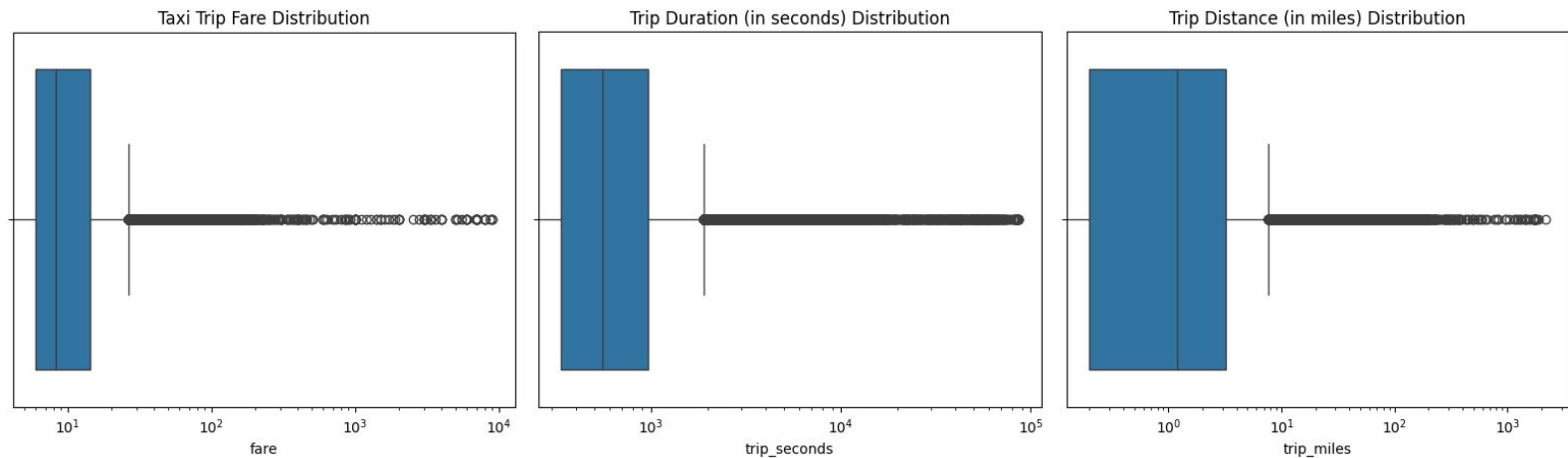
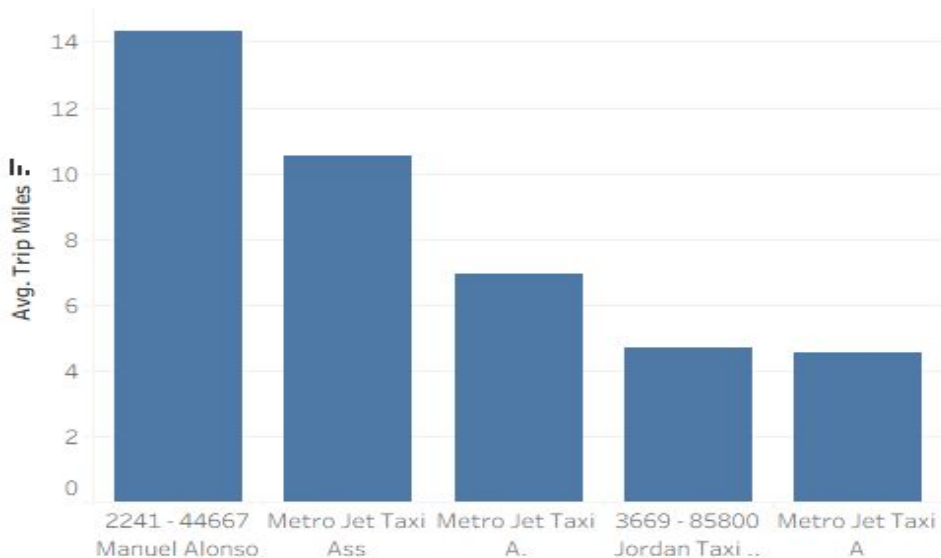




I began my exploration by checking for outliers. Immediately evident was the extreme volume of outliers visible on box-and-whisker plots. Having been forced to adjust them to a logarithmic scale made it almost seem that the outliers were actually the majority, until I took into account the observations numbered at one million. With this, I felt secure in proceeding with the next step of cleaning.

The next step of cleaning was handling duplicates and missing values. The former was completely absent, however I initially found a rather concerning amount of missing values in the locational data across the board. I thought first to clean the data up by equating the coordinates to the pickup and dropoff locations to each other, but after no change to the amount of missing details, I proceeded to the City of Chicago's website to find further information. To my surprise, the redactions were intentional for purposes of anonymity, being that Chicago is a major city with well-known persons having residence there. With this information in mind, I decided to leave the suppressed details within the data-set as outliers instead. Furthermore, due to the extreme size of the dataset with over 20 columns per observation, it felt expedient to leave things as they lay so I can attain a more-thorough analysis of the data.

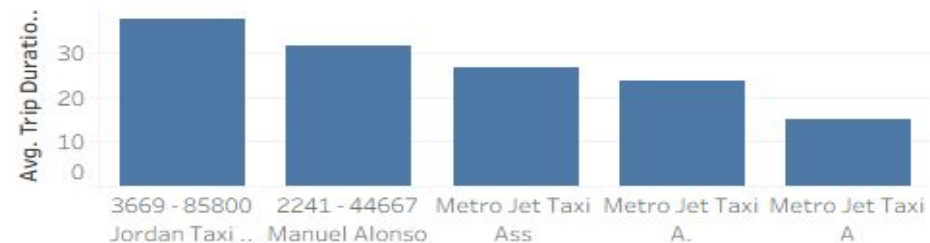
Avg Dist (Miles) by Company



Total Taxis by Company



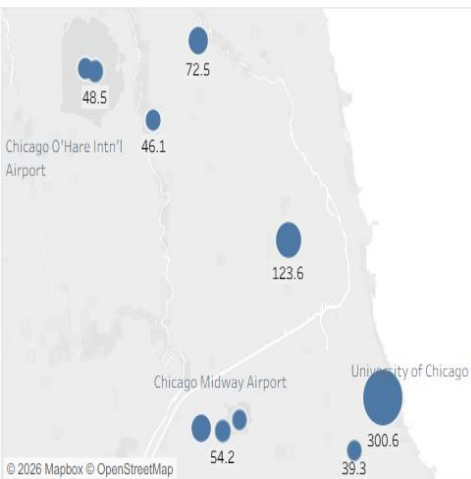
Avg Duration (Minutes) by Company



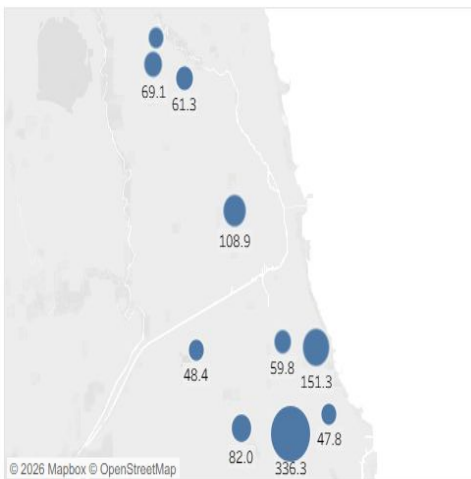
Drawing on the Top 5 Taxi companies by Average Fare, there's an interesting pattern emerging. Of the 5 companies that draw in the most fare on average, 3 of those companies appear to be the same company under different names. It may be worth investigating if there is any attempts at tax-dodging or fraud occurring with that repeating instance of slightly-different company names. The fact that "Metro Jet Taxi Association" appears to possess a starkly larger number of taxi vehicles in comparison to any other company in the Top 5 Fare-earners also supports this.

Asides from that potential-tax-dodging oddity, everything else follows an expected pattern. The top 5 used companies have a fairly standard slope with how far and how long their trips tend to last, so no I see no further concerns that could signal any other fowl-play within the taxi companies serving the city.

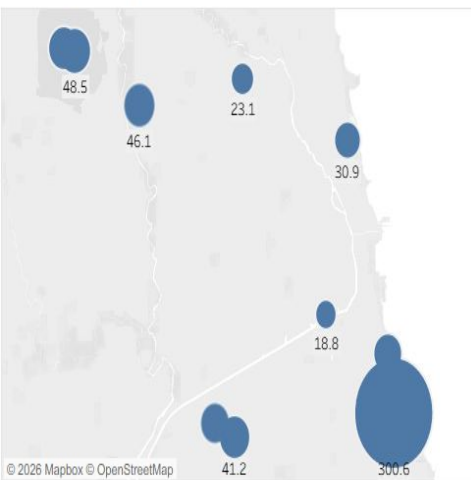
Top 10 Trip Totals by Pickup



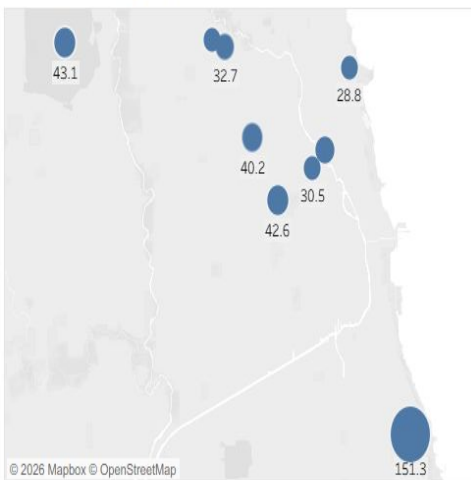
Top 10 Trip Totals by Dropoff



Top 10 Tips by Pickup



Top 10 Tips by Dropoff



From there, I moved to reviewing map coordinate trip totals and tipping habits by areas of pick-up and drop-off. It was immediately apparent that the people living in the areas in and around the University are the more likely to take the taxis, with those regions consistently appearing within the Top 10 for average amounts of tipping and trip totals. Given the fact it's one of the country's more notable private universities, it makes sense that the wealthiest riders of taxis would be the well-off students of the institution. Even with this in mind, they may still be seeking a less-costly form of transit that is less vulnerable to many common taxi practices.

The next largest averages are notably in the O'Hare International and Midway Airports. Tourists arriving to the city face a similar predicament to students of the University, in that they're unfamiliar with the lay of the city as well as what would be considered a fair price. If they were to have the option of cheaper, safer transit then this would also be a viable means of commute.

The lowest of the Top 10 would lay in the housing districts in between the airports and University, those being citizens of the city. Given that those are residents, they likely have budgets for their travel and may appreciate an alternative to taxi services.

Top 10 Earners

Company	Number of Trips	Fare	Tolls (Clean)	Tips	Extras	Trip Total
Taxi Affiliation Services	208,681	\$2,706,999.19	\$78.01	\$286,488.34	\$251,691.97	\$3,245,301.95
Flash Cab	97,392	\$1,455,640.47	\$0.00	\$129,889.67	\$86,749.50	\$1,677,092.94
Yellow Cab	54,926	\$667,808.31	\$0.00	\$65,778.13	\$44,482.00	\$778,068.44
Blue Ribbon Taxi Associat..	50,218	\$600,131.10	\$0.00	\$63,655.10	\$52,211.01	\$715,997.21
Dispatch Taxi Affiliation	48,370	\$572,713.78	\$465.69	\$56,369.21	\$45,521.16	\$675,069.84
Chicago Carriage Cab Corp	38,386	\$547,338.76	\$0.00	\$67,365.52	\$48,693.00	\$672,261.43
Sun Taxi	31,663	\$521,735.71	\$176.66	\$76,109.56	\$48,583.73	\$654,898.11
Choice Taxi Association	38,871	\$525,850.38	\$284.36	\$65,066.66	\$44,839.52	\$636,040.92
City Service	31,493	\$496,682.43	\$54.80	\$66,995.36	\$40,608.05	\$612,257.46
Medallion Leasin	24,534	\$391,515.90	\$260.21	\$47,585.23	\$35,881.63	\$481,365.95

Bottom 10 Earners

Company	Number of Trips	Fare	Tolls (Clean)	Tips	Extras	Trip Total
3201 - C & D Cab Co Inc	3	\$32.15	\$0.00	\$0.00	\$0.00	\$32.15
585 - Valley Cab Co	3	\$28.95	\$0.00	\$0.00	\$1.00	\$29.95
3094 - G.L.B. Cab Co	3	\$27.75	\$0.00	\$0.00	\$2.00	\$29.75
3253 - Gaither Cab Co.	1	\$11.85	\$0.00	\$2.55	\$1.00	\$15.40
2092 - Sbeih company	1	\$14.45	\$0.00	\$0.00	\$0.00	\$14.45
4197 - Royal Star	1	\$8.05	\$0.00	\$2.00	\$0.00	\$10.05
2823 - Seung Lee	1	\$8.25	\$0.00	\$0.00	\$0.00	\$8.25
3591- Chuk's Cab	1	\$5.45	\$0.00	\$0.00	\$1.00	\$6.45
3385 - Eman Cab	1	\$5.85	\$0.00	\$0.00	\$0.00	\$5.85
2809 - 95474 C&D Cab Co l..	1	\$5.45	\$0.00	\$0.00	\$0.00	\$5.45

Lastly, to gain a full picture of the greater taxi service, I organized them by the sums of their trip totals to compare their total performance. The previously-mentioned "Metro Jet Taxi Association" doesn't even appear within the top 10 of total earnings, with the Taxi Affiliation Services displaying a strong lead in both earnings and numbers of trips. Of those in the top 10, 6 of them indicate having needed to utilize Tollways, showing a clear need for some alternative means of transport that could also ease up traffic burdens on the highways.

Of those in the bottom 10, all are companies that didn't even get beyond single-digit numbers of trips for the total observations. They may prove a good source to go to for employment should the motion to improve public transit go forward. Any of those businesses that may have shuttered after a failed start-up may also help revitalize the commercial faith of those looking to start their own career or private business within the limits of the city.